SELVAKUMAR SADURSHAN

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Summary

Aspiring Machine Learning Engineer passionate about applying Data Science and AI to solve real-world problems. Skilled in Python, Machine Learning, Deep Learning, data analysis, SQL, REST APIs, and Git. Familiar with Generative AI, LLMs, Prompt Engineering, and RAG. Experienced with Pandas, NumPy, Scikit-learn, TensorFlow, and PyTorch. Knowledgeable in AWS cloud services and continuously learning MLOps, LLMOps, and Agentic AI to develop scalable AI solutions

Professional Experience

Intern - Data analyst, *HSBC* **Aug 2025 – Feb 2026**

- Performed data extraction, cleaning, validation, and customer data matching using Python and Excel.
- Developed Power BI/Excel dashboards and reports to analyze department performance and work quality.
- Automated email workflows and customer identification processes using Excel Macros and Python, improving operational efficiency.

Student Counselor *PrimeleedManagement* **Jun 2022 – Oct 2022**

- Served as the main point of contact for UK clients and universities, ensuring clear communication and strong relationships.
- Guided students on course selection, university admissions, and application processes.

Course-Consultant Officer *Eclub* **Aug 2021 – Feb 2022**

- Conducted workshops to explain e-club courses in an interactive way.
- Helped students gain practical, hands-on skills through these sessions.

Education

Bachelor's Degree in Data-Science, *SLIIT* 2022 feb – Present

Certificate level of BCS Higher Education Qualification (HEQ) *Live College* 2023 mar - 2023 Sep
BCS, The Chartered Institute for IT : 1st Stage Completed

Diploma In IOT *Esoft Metro Campus* 2019 jan - 2019 june
Basics Of IOT

Skills

Programming Languages
R, C, Java, Python, C++, kotlin , React-Native , Java- Script

Mobile Development
Android Application Development

Data-warehousing
ETL , SQL , Data cleaning , Data Transformation

AILibraries
Tensor-flow, Skilearn, kears, Pytorch

Web Development
MERN Stack, Spring-Boot , django

UI/UX Design
UI Designing, Figma, Draw.io

Databases:
Superbase, PostgreSQL, Firebase , Mongodb

Tools
Power-BI, Trello, Kanban board , Docker

Certifications

- Introduction to Party Rock : AWS Educate
- Prompt Engineering : Greating Learning
- Introduction to Generative Ai : AWS Educate
- IBM data-Analyst : Udemy
- IBM Machine Learning With Python : Coursera
- Advanced ML Algorithms & Unsupervised Learning : Packt

Projects

Anomaly and Health Monitering System On Turbo fan engine of aircraft

- Developed an Explainable AI-based health monitoring and anomaly detection system using NASA C-MAPSS turbofan data.
- Implemented LSTM digital twin and ensemble anomaly detection using GMM, Isolation Forest, LSTM autoencoder, and K-Means.
- Performed time-series processing, feature engineering, root-cause analysis, and anomaly visualisation using Python.

Edge Defect Segmentation

- Developed an industrial surface-defect segmentation model using KolektorSDD2, targeting edge deployment under a 50 ms P95 CPU latency constraint.
- Applied knowledge distillation, structured pruning, INT8 quantization, and mixed-precision optimization to reduce model cost while preserving segmentation performance.
- Conducted multi-configuration CPU benchmarking, per-layer sensitivity analysis, and quantization ablation to identify the optimal deployment configuration.

hybrid-rag-assistant

- Built a RAG-based document assistant using FAISS, BM25, MiniLM embeddings, and cross-encoder reranking.
- Implemented grounded AI responses with page-level citations, conversational memory, and hallucination/refusal control.
- Developed an end-to-end FastAPI + React application with document upload, retrieval, streaming, and structured extraction.
- Achieved 1.00 Hit@1 and 1.00 MRR through retrieval evaluation and optimization.

Climate Policy Analysis Agent

- Built a multi-document NLP system for semantic document comparison using embeddings and Hungarian matching for sentence-level alignment.
- Implemented zero-shot/fine-tuned classification with confidence scoring and structured extraction of quantitative targets such as percentages, dates, and monetary values.
- Developed a RAG pipeline using chunking, vector search, and map-reduce summarization with page-level citations for grounded responses.
- Built an evaluation framework with gold-labeled test data and precision/recall/F1 metrics, deployed using Docker Compose, async job processing, and CI/CD.

IOT Parking System-

- Developed a comprehensive ML pipeline for parking occupancy analysis, featuring temporal trend modeling, anomaly detection, and data clustering.
- Engineered and trained compact TinyML models, exporting artifacts to TensorFlow Lite for optimized, low-latency edge inference.
- Built predictive occupancy models including status classifiers and regression networks to forecast real-time parking space availability.
- Integrated the ML architecture with a Node.js backend and React frontend, generating automated temporal visualizations for pattern analysis.